

Batched recursion for partial-swap frames

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Setting. In the two-stage bit motif (Paurel's fork of `CrocSwap/integer-mult-bounds`, `notes/two-stage-construction.tex` and `notes/direct-swap-transfer.tex`), frames are partial swaps

$$S_p = \begin{pmatrix} I - p & p \\ p & I - p \end{pmatrix}$$

on the paired fields (H, D) , where p is an idempotent of rank a in m coordinates. A frame change along an edge is a partial swap $S_{V \oplus U}$. In the recursion each lower-lower Bruhat transposition is one recursive interchange call. Batched recursion (`CrocSwap/integer-mult-bounds` PR #10, `notes/projector-batching.tex`) charges an order-preserving run of t aligned transpositions as a single interchange of width tb . Below we show that the same batching applies to partial-swap frames.

Lemma 1 (Partial-swap factorization). *Let p be an idempotent of rank a with $m/2 < a < m$. Put $k = m - a$, $t = 2a - m$, and split the coordinates into $L = [1, k]$, $M = [k + 1, m - k]$ and $R = [m - k + 1, m]$. Suppose the corner $p_{L,R}$ is invertible. Then the lower-lower Bruhat factorization $S_p = E_1 \Pi E_2$, with respect to the field order $H_1, \dots, H_m, D_1, \dots, D_m$, has exactly a nontrivial transpositions:*

- k transpositions $H_l \leftrightarrow D_{\sigma(l)}$ for $l \in L$, where $\sigma : L \rightarrow R$ is the Bruhat permutation of the corner $p_{L,R}$;
- the t aligned transpositions $H_i \leftrightarrow D_i$ for $i \in M$, an order-preserving run that forms a single interchange of width tb .

In particular the rank sum is unchanged, since $k + t = a$.

Proof. Write $q = I - p$, an idempotent of rank k , as $q = UV$ with $VU = I_k$. Then $p_{L,R} = -U_L V_R$, so U_L and V_R are invertible. The Bruhat permutation

w of S_p is determined by the ranks of its blocks of rows $1..i$ and columns $j..2m$.

H rows against D columns. This block is p itself, so these pivots of S_p are the pivots of p . Eliminate the invertible corner $p_{L,R}$ by lower row and column operations. On rows $M \cup R$ and columns $L \cup M$ the Schur complement is

$$\begin{pmatrix} -U_M U_L^{-1} & I_t \\ Z & -V_R^{-1} V_M \end{pmatrix},$$

which has rank $a - k = t$. Its block I_t places these pivots at (i, i) for $i \in M$. So the pivots of p are σ on $L \rightarrow R$ together with the diagonal of M .

Symmetry. S_p is an involution. Inverting $S_p = E_1 w E_2$ gives a lower-lower factorization with w^{-1} , so w is an involution, and it also has the pivots $D_{\sigma(l)} \rightarrow H_l$ and $D_i \rightarrow H_i$.

Remaining rows. The rows H_R have no pivot in D_L , because p has no pivot column in L . For rows $H_{\leq i}$ and columns $H_{\geq j} \cup D$, the rank is $|[j, i]| + \text{rank } p[1.. \min(i, j-1), \cdot]$. On R this equals the rank function of the identity. The ranks of the blocks $(H \cup D_{\leq i'}) \times D_{\geq j'}$ give the same for D_L . So H_R and D_L are fixed points of w .

The diagonal factors are units, so every transposition costs one interchange of its width. $\square$

Remark (Corners). For the frames of the two-stage motif the corners are mostly singular in the original coordinates. One fixed unimodular change of basis S_0 , as in PR #10's controlled-basis lemma, makes them all invertible. PR #10's simultaneous-conjugation lemma applies unchanged: conjugating by $\text{diag}(S_0, S_0)$ preserves the identity, the full swap, the endpoint correction $PF = E$, and the composition rule for orthogonal summands.

Existence of S_0 at every h . After conjugation by $S \in SL_m$ the corner of an edge is $-(S^{-1}U)_L(VS)_R$. Since U has full column rank and V full row rank, each of $\det(S^{-1}U)_L$ and $\det(VS)_R$ is a polynomial in the entries of S (with $S^{-1} = \text{adj } S$) that is not identically zero on SL_m : some S sends the columns of U to e_L , and some S sends the rows of V to e_R^T , each with determinant fixed to 1 by scaling a column outside the image. The motif has finitely many edge classes, so the product of these polynomials over all classes is a nonzero polynomial on the irreducible variety SL_m . $SL_m(\mathbb{Z})$ is Zariski dense in SL_m , so some integral unimodular S_0 makes every corner invertible. The checks below find one explicitly at $h = 6, 7, 8$.

Choice of prime. The factors E_1, E_2 have rational entries whose denominators divide products of corner minors, and S_0^{-1} is integral. These are fixed for a fixed motif, so they add finitely many primes to the list the odd radix q of the two-stage transfer must avoid, alongside the denominators of its own triangular factors.

Use in the recursion. PR #10's integer-width recurrence needs, for each edge, only a list of child widths each at most $m - 1$, with a full-rank edge contributing m singletons. A selected edge contributes h singleton calls and one aligned block of width $m - 2h$; the copy-correction child of the two-stage transfer has width 1; every other edge is compiled as in the two-stage motif. Row merging and parking are unchanged, since they act on the list of children and not on the frames.

Selected edges and saving. In the two-stage bit motif at $h = 32$ ($m = 1024$) every auxiliary role, side or centre, in either stage, has exactly one edge of rank $m - h = 992$. In stage one it is the final enlargement $D_1 \otimes Q \rightarrow F$, and in stage two it is the initial growth $0 \rightarrow D_0$. There are $2v(R + c_0)$ such edges, and each is compiled as h singleton calls plus one block of width $t = m - 2h = 960$. With the counts of `certificates/two-stage.json` ($v = 4960$, $R = 123157$), the batched moment certifies

$$a_b = \frac{22157}{5 \cdot 10^9} \approx 4.43 \cdot 10^{-6},$$

against a true root of $4.4315 \cdot 10^{-6}$. That is 9.3 times the unbatched saving $4.7 \cdot 10^{-7}$.

Checks. `independent/partial-swap/` rebuilds $E_1 \amalg E_2 = S_p$ exactly and verifies triangularity and the pivot profile. It covers 270 random and sparse idempotents with $3 \leq m \leq 12$ and every $a \in (m/2, m)$, all 169 cases with invertible corners matching. When the corner is singular the profile fails, which shows the conjugation is needed. For the two-stage frames at $h = 6, 7, 8$ it checks:

- monotonicity on X_a : $U_a \subset t_1 \otimes t_2 \subset F \otimes t_2 \subset (t_1^\perp \otimes F) \perp (t_1 \otimes t_2) \subset F$;
- the rank sums m per side role, $m + 2h$ per centre role and $m - 1$ per data role, which reproduce $s = Wm - 2N + 2L$;
- the selected edge classes and their Bruhat profiles after conjugation by one unimodular S_0 : 78/78, 20/20 and 10/10.